

Notes on revisions made to "Building Classification Using a YOLO Object Detection Model Trained on Aerial Imagery"

The authors would like to thank the editor and reviewers for their constructive comments and suggestions on the paper titled *Building Classification Using a YOLO Object Detection Model Trained on Aerial Imagery* that have helped improve the quality of this manuscript. The resubmitted manuscript has undergone a thorough revision according to the reviewers' comments. Please see our responses below.

Reviewer A

Reviewer Comment A.1 — I recommend asking an editor to review for grammar.

Reply: Thank you for this feedback. We have revised our paper and fixed the grammatical errors.

Reviewer Comment A.2 — As a reader, I am interested to know how often traffic engineers create travel forecasting models for areas without available data. Perhaps you could include an example of a project that resulted in a severely over or undersized project.

Reply: Thank you for this suggestion. We have added information about bias travel forecasting models and model transferability to the end of the first paragraph in the introduction.

Reviewer B

Reviewer Comment B.1 — Revise the paper to remove conversational phrasing and use a formal academic tone throughout. Avoid first-person pronouns (e.g., "I, and we").

Reply: We appreciate the reviewer's feedback. The standard practice in travel demand modeling literature has been to use first person pronouns. Since the official IEEE paper guidelines do not prohibit their use, we did not make any changes.

Reviewer Comment B.2 — Include specific metrics in the discussion of results instead of phrases such as "almost 100 percent of the time," for better clarity in the data analysis.

Reply: Thank you for this comment. We have updated this section to include more specific metrics and removed phrases that were more vague.

Reviewer Comment B.3 — How is the model performance presented in the paper. Are there other more metrics that can be used for clarity and better readability.

Reply: Thank you for your feedback. The model’s performance is measured by it’s ability to recognize the building types it was trained on, which is best displayed by the confusion matrices given in the paper. The model does also provided individual confidence levels for each building it detected (as can be seen in Figure 2), but discussing these results was beyond the scope of this paper. In depth discussion about the technical side of the model training was also beyond its scope.

Reviewer Comment B.4 — In the conclusions section, the sentences about identifying other attributes of occupants by the presence and type of vehicles, children’s toys, and other objects as indicators of income or the presence of children may raise privacy concerns that should be addressed.

Reply: Thank you for your feedback. This was discussed in the last paragraph of the paper talking about limitations. We have moved this short discussion to be directly after the sentences discussing these attributes.

Reviewer Comment B.5 — Additionally, revise the paper for grammar issues and typos throughout. Examples: “satallite”, “socioeconomic”.

Reply: Thank you for this feedback. We have revised our paper and fixed the grammatical errors and typos.